

Ricardo (Zhicong) Xie

Simulation / Analysis Intern | FEA, CFD and Validation

Auckland, New Zealand | 02902729390 | zxie588@aucklanduni.ac.nz | Portfolio: <https://zxie588-alt.github.io> | GitHub: <https://github.com/zxie588-alt>

Valid NZ student visa | No sponsorship required | Full-time 17 Nov 2026 - late Feb 2027

PROFILE

- Mechanical engineering master's student specialising in structural FEA and thermal-fluid CFD, with project work in Abaqus/Standard, ANSYS Fluent/CFX and model credibility checks.
- I support engineering teams by delivering analysis with clear load cases, boundary conditions, mesh checks, mass/heat balance review, result interpretation and documented model limitations.
- Project screenshots and source links: <https://zxie588-alt.github.io>

WORK RIGHTS AND AVAILABILITY

- Valid NZ student visa. No sponsorship required.
- Full-time available: 17 Nov 2026 - late Feb 2027 (university break).
- Part-time eligible: up to 25 hours/week during semester. Auckland based; open to relocation for suitable roles.

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Current study and project work in building services, smart manufacturing, CFD/thermal modelling, composite structures and engineering methods.

Relevant coursework: Finite Element Analysis, Computational Fluid Dynamics, Composite Materials, Thermal Modelling

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PROJECT WORK

Composite Formula SAE Front Wing Analysis

Abaqus/Standard, shell FEA, composite layout, structural checks

- Developed a shell FEA model of a composite Formula SAE front wing in Abaqus/Standard using S4R elements, layered laminate definitions and local mesh refinement at high-stress attachment points.
- Converted aero downforce/drag, lateral cornering and 4.1 G bump conditions into structural load cases for strength and stiffness checks.
- Evaluated ply-level stress utilisation against fatigue and ultimate failure criteria, achieving about 74% peak utilisation under the governing 4.1 G bump case with SF = 1.5.

Solar Chimney CFD Reproduction

ANSYS Fluent/CFX, conjugate heat transfer, natural convection

- Built a CFD reproduction workflow for a laboratory-scale solar chimney with coupled air, absorber and cover domains.
- Validated model credibility through mesh-sensitivity checks, global mass/heat balance review, Reynolds/Rayleigh scaling and solver-behaviour comparison before trusting velocity results.
- Documented model boundaries, limitations and recommended physical data points needed for future experimental correlation.

Ruggedized Outdoor IoT Sensor Module

SolidWorks, enclosure design, DFM notes, Keil C, concept validation

- Designed a SolidWorks enclosure concept with screw bosses, gasket stack-up, PCB and battery placeholders, cable/strap mounts and service access.
- Applied DFM principles to improve enclosure features for CNC machining discussion and 3D-print prototype planning.
- Added FEA-style screening for corner load, strap lug and screw boss regions; kept an open test list for sealing, impact, thermal exposure and firmware validation.
- Built concept firmware in Keil C with mock sensor readout, battery monitoring and a simple module state machine; documented the electronics and firmware architecture in the portfolio.

NZ Residential MVHR / ERV Ventilation Concept

SolidWorks, AutoCAD/DXF, HVAC sizing, NZBC G4/H1 context, commissioning plan

- Developed a New Zealand residential MVHR/ERV ventilation concept aligned with local building-code requirements.
- Modelled the MVHR unit, supply/extract/outdoor/exhaust ducts, diffusers, grilles and hangers in SolidWorks; exported DXF, STEP, STL, PNG and a component list for review.
- Sized the system for 60 L/s balanced supply/extract in normal operation and 85 L/s boost extract mode, calculating external static pressure targets of about 80 Pa and 120 Pa.
- Checked the design against NZBC G4/H1 and Healthy Homes ventilation context; developed a commissioning test plan covering CO₂, humidity, airflow and fan-power measurements.

Orbital AI Data Centre Feasibility Study - Systems Lead

Stakeholders, requirements, ConOps, trade studies, risk, verification

- Led the systems-engineering workstream for a 5-person team concept covering a 24-satellite LEO AI data-centre architecture.
- Set up the stakeholder map, requirements hierarchy, ConOps, architecture options, trade-study criteria, risk register and verification gates.
- Delivered a feasibility-report structure with trade-off matrices, risk logic and verification-gate framework across orbit/coverage, power, thermal rejection, communications, launch mass, cost and sustainability.

TECHNICAL SKILLS

- Proficient in: Abaqus/Standard, ANSYS Fluent/CFX, shell FEA, CFD, conjugate heat transfer and natural-convection modelling
- Proficient in: load-case definition, boundary conditions, mesh-sensitivity checks, assumptions log, result interpretation and model-limit documentation
- Familiar with: composite failure criteria, ply utilisation checks, Rayleigh/Reynolds scaling, global mass/heat balance validation

LEADERSHIP AND COMMUNICATION

- Led team-level integration in the orbital data-centre project and commonly take the organiser role in group work.
- Skilled in technical visual communication and structured documentation; experienced in presenting engineering analysis clearly and producing evidence-based project deliverables.
- Professional working proficiency in English; native Mandarin Chinese.